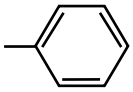
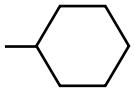
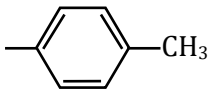
RACE
Regular Analysis Through Continuous Exercise

Electronic Displacement Effect : RACE-01

Inductive effect (Groups showing +I and -I effects)

1. Which of the following groups have + I effect :

(A)  (B)  (C)  (D) $-CH=CH_2$

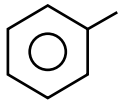
CED089

2. Which of the following is false order of – I effect ?

(A) $-F > -Cl > -Br > -I$

(B) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2$

(C) $-F > -OH > -NH_2$

(D)  $> -C \equiv CH > -H$

CED102

3. What is the correct order of + I effect ?

(A) $-O^{\ominus} > -CH_3 > -CMe_3$

(B) $-CO_2^{\ominus} > -O^{\ominus} > -CMe_3$

(C) $-O^{\ominus} > -CH_2Me > -D > -H$

(D) None

CED103

4. Which of the following groups have –I effect :

(A) $-OH$ (B) $-\overset{\overset{O}{\|}}{C}-OH$ (C) $-CH_3$ (D) $-OCH_3$

CED107

5. Which of the following group (s) exerts +I effect ?

(A) $-CO_2^{\ominus}$ (B) $-CH_3$ (C) $-NH_3^{\oplus}$ (D) $-OH$

CED245

Stability of Intermediates

6. Most stable carbanion is :-

(A) $HC \equiv C^{\ominus}$ (B) $C_6H_5^{\ominus}$ (C) $(CH_3)_3C-CH_2^{\ominus}$ (D) $(CH_3)_2C=CH^{\ominus}$

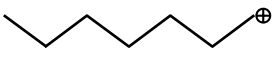
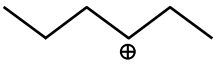
CED090

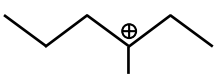
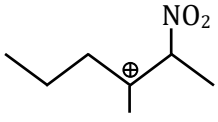
7. Which of the following is most stable carbocation ?

(A) $\overset{\oplus}{C}H_3$ (B) $CH_3-\overset{\oplus}{C}H-CH_3$ (C) $CH_3-\overset{\oplus}{C}H_2$ (D) $CH_3-\overset{\oplus}{\underset{\underset{CH}{|}}{C}}-CH_3$

CED104

8. The most stable carbocation among the following :

(A)  (B) 

(C)  (D) 

CED105

Ranker's Academy JEE
For More Join: @IITJEE_Advanced

9. Give stability in increasing order of following intermediates:

(a) $CH_3-\overset{\oplus}{C}H_2$ (b) $CH_2=\overset{\oplus}{C}H$ (c) $CH\equiv\overset{\oplus}{C}H$

(A) a > b > c (B) c > b > a (C) c > a > b (D) b > a > c

CED106

Electrophiles and Nucleophiles

10. Which of the following species is not electrophilic in nature :-

(A) BH_3 (B) $H_3O^{\oplus}$ (C) $^{\oplus}NO_2$ (D) $Cl^{\oplus}$

CED096

11. Wrong statement regarding methyl carbonium ion $\overset{\oplus}{C}H_3$.

(A) It is sp^2 hybridised

(B) Vacant orbital is sp^2 hybridised

(C) Vacant orbital is perpendicular to molecular planar and in pure p-orbital

(D) It is electrophile with sextet of electron

CED101

12. (a) $CH_2=CH_2$ (b) $BH_4^{\ominus}$ (c)  (d) NH_3 (e) Me_2S

Above are examples of some Nucleophiles if

X = Number of compounds in which π bonding orbital is HOMO.

Y = Number of compounds in which σ bonding orbital is HOMO.

Z = Number of compounds in which Non-bonding (L.P) is HOMO.

Then X + Y – Z = ?

Pankers Academy JEE

CED364

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	C	ABD	AB	A	D	C	A	B
Question	11	12								
Answer	B	1								

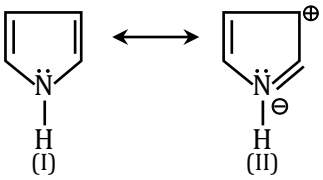
For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Electronic Displacement Effect : RACE-02

Resonating structures and their Stabilities

1. Examine the following two structures for pyrrole and choose the correct statement given below



(A) II is not an acceptable resonating structure because carbonium ions is less stable than nitride ion

(B) II is not an acceptable resonating structure because there is charge separation

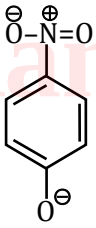
(C) II is not an acceptable resonating structure because nitrogen has ten valence electrons

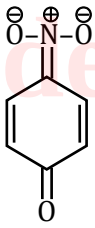
(D) II is an acceptable resonating structure

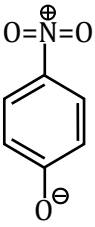
CED118

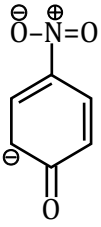
2. The most unlikely representation of resonance structure of p-nitrophenoxide ion is-

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(A) 

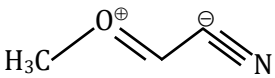
(B) 

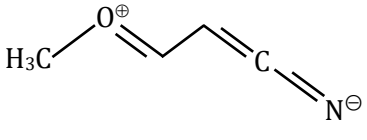
(C) 

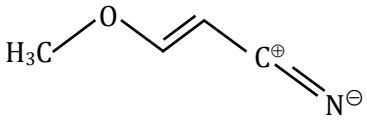
(D) 

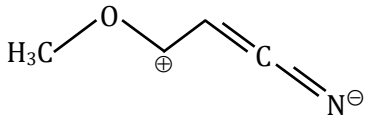
CED120

3. Which of the following structure is the most stable ?

(A) 

(B) 

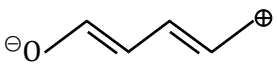
(C) 

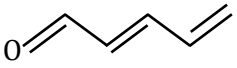
(D) 

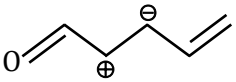
CED125

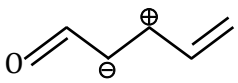
For More Join: @IITJEE_Advanced

4. The least stable resonating structure is

(A) 

(B) 

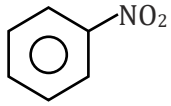
(C) 

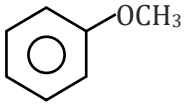
(D) 

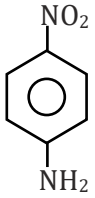
CED126

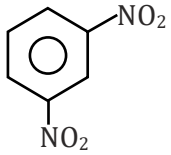
Mesomeric effect (Groups showing +M and -M effects)

5. In which of the following molecules π – electron density in ring is minimum:

(A) 

(B) 

(C) 

(D) 

CED133

6. Match the column I with column II.

	Column-I (Group attached with benzene ring)		**Column-II (Effect shown by the group)**
(A)	$-NO_2$	(P)	–R effect
(B)	$-O^{\ominus}$	(Q)	+R effect
(C)	$-O-CH_3$	(R)	+I effect
(D)	$-C\equiv N$	(S)	–I effect

(A) (A) → P, S, (B) → Q, R, (C) → Q, S, (D) → P, S

(B) (A) → P, Q, (B) → Q, R, (C) → R, S, (D) → P, Q

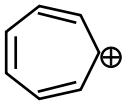
(C) (A) → P, S, (B) → Q, R, (C) → Q, S, (D) → P

(D) (A) → P, S, (B) → Q, R, (C) →S, (D) → P, S

CED138

Aromaticity

7. Among the following the most stable cation is

(A) 

(B) $Ph-\overset{\oplus}{C}H_2$ (C) $(CH_3)_3C^{\oplus}$ (D) $CH_2=CH^{\oplus}$

CED150

8. STATEMENT–I : Tropylium cation is more stable than $(CH_3)_3C^{\oplus}$

STATEMENT–II : Tropylium cation has aromatic character

(A) Statement–I is true, Statement–II is true; Statement–II is a correct explanation for statement–I

(B) Statement–I is true, Statement–II is true; Statement–II is not a correct explanation for statement–I

(C) Statement–I is true, Statement–II is false

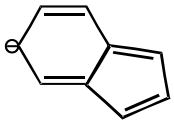
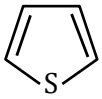
(D) Statement–I is false, Statement–II is true

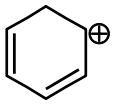
CED151

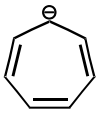
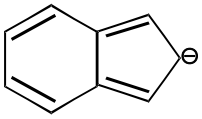
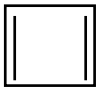
Rankers Academy JEE

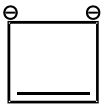
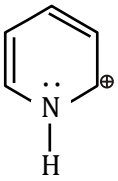
For More Join: @IITJEE_Advanced

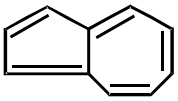
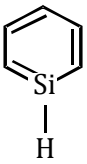
9. Select the set containing all aromatic species.

(A) (a)  (b) 

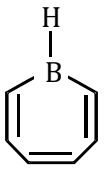
(c) 

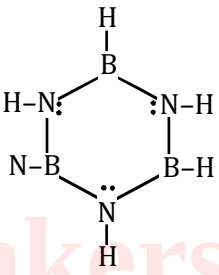
(B) (a)  (b)  (c)  (d) 

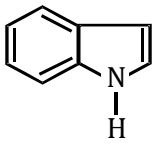
(C) (a)  (b)  (c)  (d) 

(D) (a)  (b) 

(c) 

(d) 

(e) 

(f) 

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CED165

10. Non–aromatic compound is/are ?

(A)  (B)  (C)  (D) All of these

CED166

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	C	B	C	D	A	A	A	D	AB

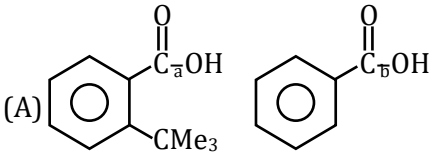
For More Join: @IITJEE_Advanced

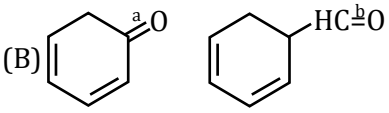
Regular Analysis Through Continuous Exercise

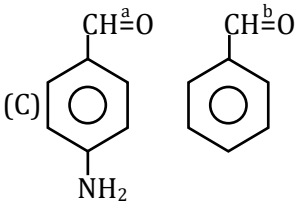
Electronic Displacement Effect : RACE-03

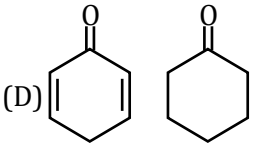
Bond length and Bond Strength

1. In which case, C – O bond length is shorter for 1st compound :

(A) 

(B) 

(C) 

(D) 

CED175

2. Which of the following is incorrect order of bond length

(A) C – C > C = C > C≡C

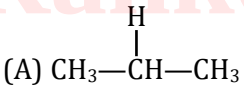
(B) C = C > C = N > C =O

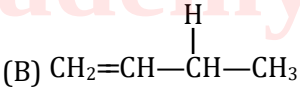
(C) C = N > C =O > C = C

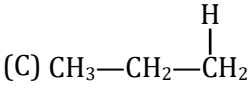
(D) C – C > C=C > C ≡ C > C–H

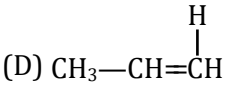
CED183

3. Which of the following C–H bonds shown would have the smallest bond dissociation energy?

(A) 

(B) 

(C) 

(D) 

CED240

4. Select correct statement :

(A) Carbon-oxygen bonds are of equal length in acetate ion

(B) Resonating structures of acetate ion are equivalent

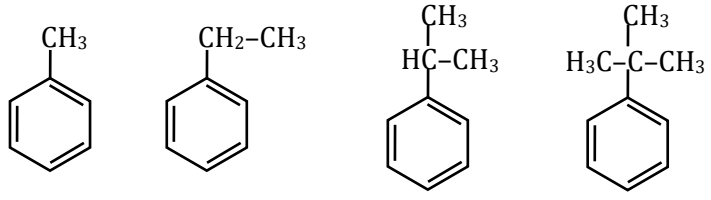
(C) Carbon-oxygen bonds are of unequal length in formate ion

(D) Resonating structures of formate ion are equivalent

CED188

Applications of Hyperconjugation

5. Correct order of electron density in the benzene ring is :



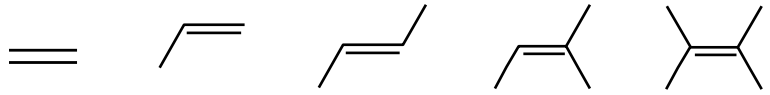
(i) (ii) (iii) (iv)

(A) i > iii > iv (B) i > ii > iii > iv (C) iv > iii > ii > i (D) iv > iii > i > ii

CED194

For More Join: @IITJEE_Advanced

6. Give increasing order of heat of hydrogenation:



(a) (b) (c) (d) (e)

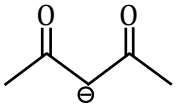
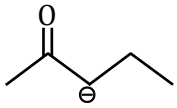
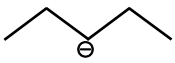
(A) a > c > d > e > b (B) a > d > c > e > b

(C) a > b > c > d > e (D) b > a > c > d > e

CED205

Stability of Intermediates

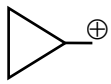
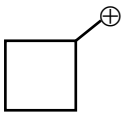
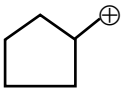
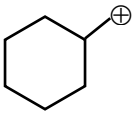
7. Correct order of intermediates in increasing order of their stability.

(a)  (b)  (c) 

(A) a > b > c (B) c > b > a (C) b > a > c (D) a > c > b

CED197

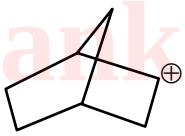
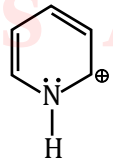
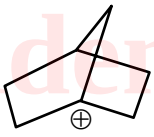
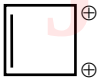
8. Among given cations which is most stable

(A)  (B)  (C)  (D) 

CED198

9. Among the following, the least stable cation is :

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(A)  (B)  (C)  (D) 

CED200

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	A	C	B	ABD	B	C	A	A	C

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electronic Displacement Effect : RACE-04

Acidic strength

1. Arrange acidity of given compounds in decreasing order:

(I) $CH_3–NH–CH_2–CH_2–OH$ (II) $CH_3–NH–CH_2–CH_2–CH_2–OH$

(III) $(CH_3)_3\overset{\oplus}{N}–CH_2–CH_2–OH$

(A) III > I > II (B) III > II > I (C) I > II > III (D) II > I > III

CED213

2. Select correct order regarding acidic strength of given compounds :

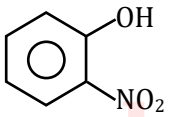
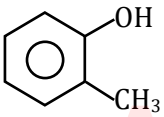
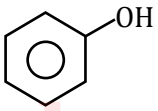
(1) o-methylbenzoic acid (2) m-methylbenzoic acid

(3) p-methylbenzoic acid (4) benzoic acid

(A) 1 > 2 > 3 > 4 (B) 4 > 3 > 2 > 1 (C) 1 > 4 > 2 > 3 (D) 3 > 2 > 4 > 1

CED216

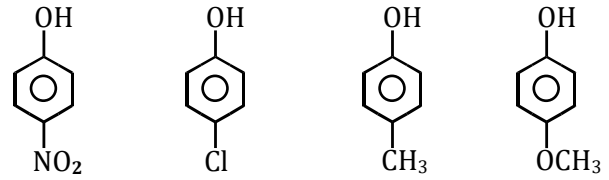
3. Which one of the following compounds is most acidic:-

(A)  (B)  (C)  (D) $ClCH_2CH_2OH$

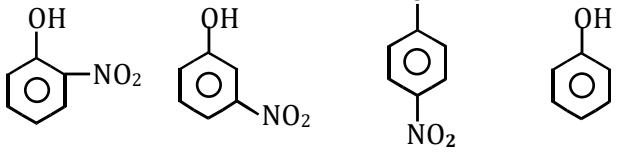
CED220

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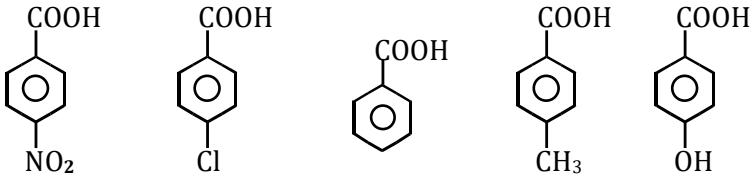
4. Correct set of acidic strength order in given option is :

(A)  (I > III > II > IV)

I II III IV

(B)  (III > I > II > IV)

I II III IV

(C)  (I > II > IV > V > III)

I II III IV V

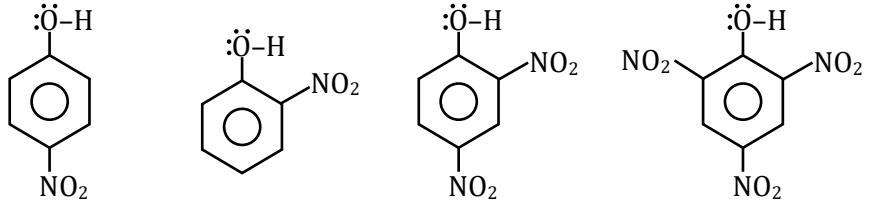
(D) H—COOH CH_3COOH $C_6H_5–COOH$ (III > I > II)

I II III

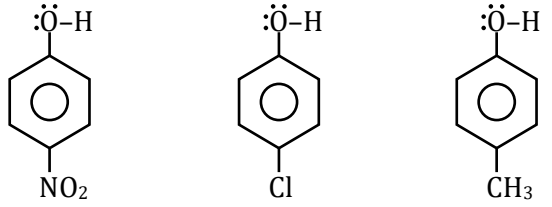
CED221

For More Join: @IITJEE_Advanced

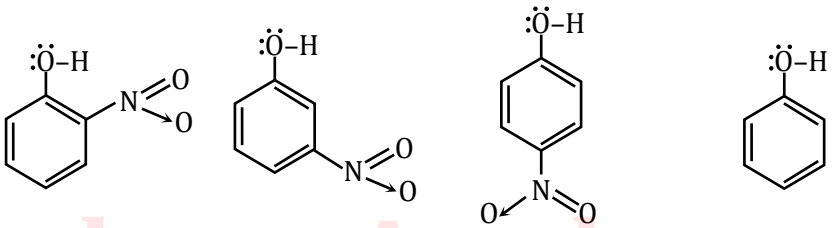
5. Incorrect set of acidic strength order in given option is :

(A) 
(a) (b) (c) (d)

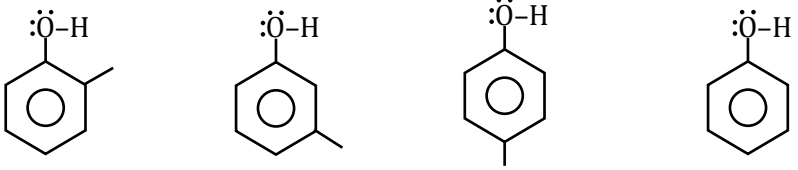
(d > c> a > b)

(B) 
(a) (b) (c)

(a > b > c)

(C) 
(a) (b) (c) (d)

(c > b > a > d)

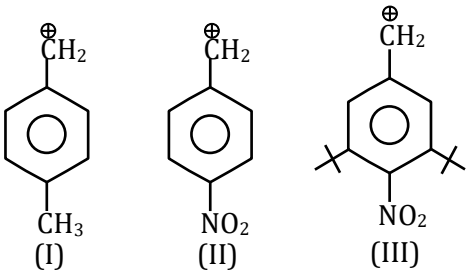
(D) 
(a) (b) (c) (d)

(d > b > c > a)

CED222

SIR Effect

6. Arrange the following carbocation in the increasing order of stability :

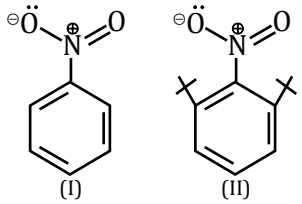

(I) (II) (III)

(A) I < II < III
(B) II < III < I
(C) III < II < I
(D) III < I < II

CED226

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For More Join: @IITJEE_Advanced

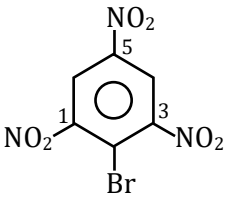
7. Consider the following two structures and choose the correct statements –



(A) carbon-nitrogen bond length structure I is greater than that in structure II
(B) carbon-nitrogen bond length in structure I is less than in structure II
(C) carbon-nitrogen bond length in both structure is same
(D) It can not be compared

CED176

8. Which of the following statements would be true about this compound :



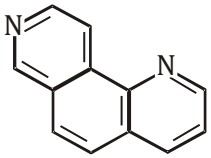
(A) All three C–N bonds are of same length.
(B) C_1–N and C_3–N bonds are of same length but shorter than C_5–N bond.
(C) C_1–N and C_3–N bonds are of same length but longer than C_5–N bond.
(D) C_1–N and C_3–N bonds are of different length but both are longer than C_5–N bond

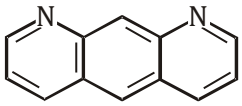
CED177

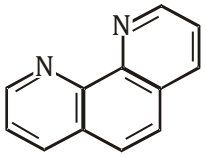
Rankers Academy JEE

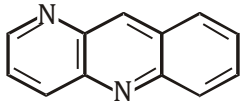
Basic strength

9. Which of the following is most basic:

(A) 

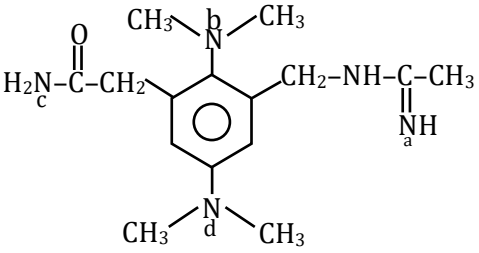
(B) 

(C) 

(D) 

CED234

10. Basicity order of N in following compound is :



(A) b > d > a > c (B) a > b > d > c (C) a > b > c > d (D) a > c > b > d

CED235

For More Join: @IITJEE_Advanced

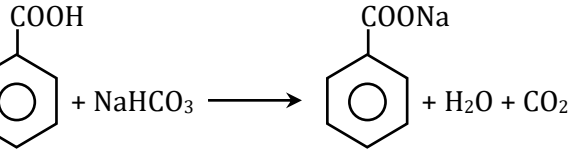
11. Which of the following is the weakest base?

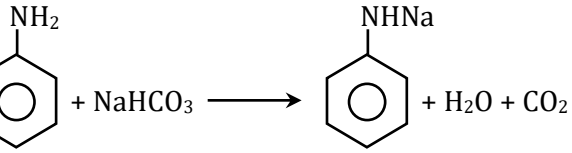
(A) $Cl^{\ominus}$ (B) $F^{\ominus}$ (C) $\overset{\ominus}{O}H$ (D) $\overset{\ominus}{C}H_3$

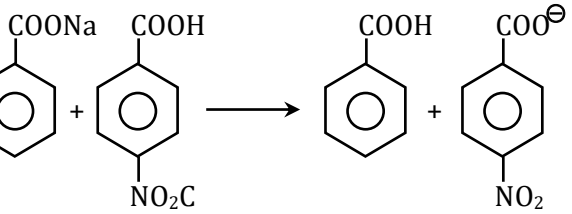
CED112

Acid Base Equilibrium

12. Identify the feasible reactions

(A) 

(B) 

(C) 

(D) Both A and C

CED239

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	A	B	C	B	B	C	C	B
Question	11	12								
Answer	A	D								

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Electronic Displacement Effect : RACE-05

Quantitative Analysis

1. In Kjeldahl's method, nitrogen present is estimated as :

(A) N_2 (B) NH_3 (C) NO_2 (D) none of these

CED365

2. The percentage of sulphur in an organic compound whose 0.32g produces 0.233g of $BaSO_4$ [At.wt.Ba =137, S=32] is

(A) 1.0 (B) 10.0 (C) 23.5 (D) 32.1

CED366

3. The desiccants used for absorbing water during Liebig's method for estimation of carbon and hydrogen are

(A) anhydrous $CaCl_2$ (B) anhydrous Na_2SO_4

(C) $Mg(ClO_4)_2$ (D)$MgSO_4$-$7H_2O$

CED367

Qualitative Analysis

4. The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of :

(A) $Fe[Fe(CN)_6]_3$ (B) $Na_3[Fe(CN)_6]$ (C) $Fe(CN)_3$ (D) $Fe_4[Fe(CN)_6]_3$

CED368

5. A compound which does not give a positive test in Lassaigne's test for nitrogen is:

(A) urea (B) hydrazine (C) azobenzene (D) phenyl hydrazine

CED369

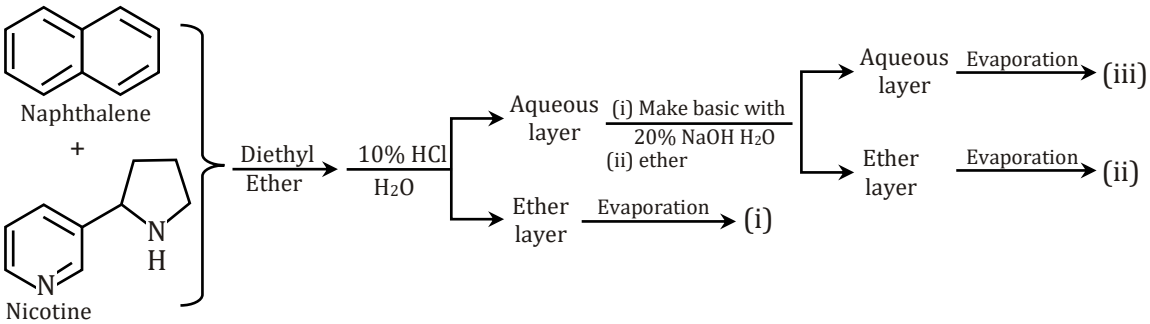
6. Disodium pentacyanonitrosyl ferrate reagent gives violet colour when which following element is present?

(A) N (B) Cl (C) F (D) S

CED370

Separation Methods

7. In the following extraction procedure, choose the number where nicotine would be found.

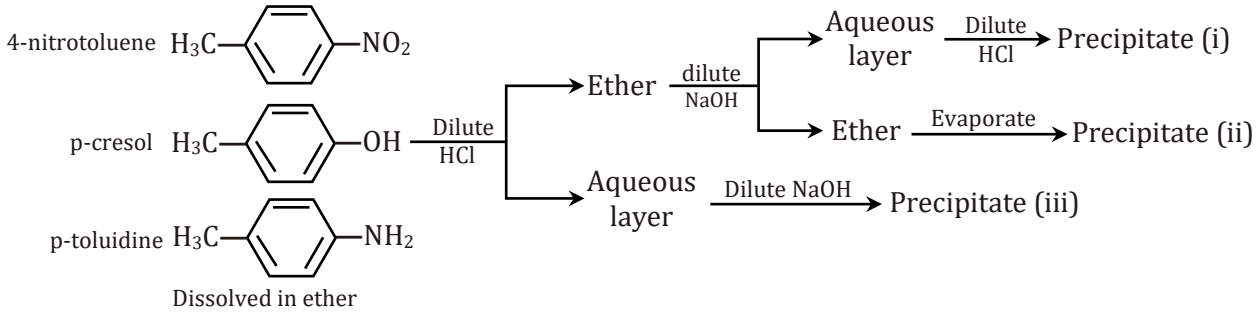


(A) i= Nicotine (B) i+iii Nicotine (C) ii-Nicotine (D) ii + iii = Nicotine

CED371

For More Join: @IITJEE_Advanced

8. Choose the answer that has the following compounds located correctly in the separation scheme.



(A) i=4-nitrotoluene, ii = p-cresol, iii = p-toluidine
(B) i=4-nitrotoluene, ii = p-toluidine, iii = p-cresol
(C) i=p-toluidine, ii = 4-nitrotoluene, iii = p-cresol
(D) i=p-cresol, ii = 4-nitrotoluene, iii = p-toluidine

CED372

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	B	B	AC	D	B	D	C	D

For More Join: @IITJEE_Advanced